

Experiment 03: Uninformed Search

Learning Objective: Student should be able to solve a given problem using uninformed search technique.

Tools: Python under Windows or Linux environment

Theory: Study and implement DFIDS Uninformed Search techniques.

Depth First Iterative Deepening Search: Choosing depth bound provides incomplete solution or admits poor solution. This variation is complete and finds best solution.

ALGORITHM:

Depth-First Iterative Deepening (DFID) Search is a search algorithm that combines the advantages of Depth-First Search (DFS) and Breadth-First Search (BFS). It repeatedly performs DFS with increasing depth limits (0, 1, 2, 3, ...). If the goal is not found within the current limit, the depth limit is increased and the search is repeated. DFID uses very little memory like DFS while still guaranteeing the shortest solution in an unweighted graph like BFS. It is complete and optimal for problems with uniform step costs, making it useful in AI search applications.

ADVANTAGES:

1.
2.

DISADVANTAGES:

1.
2.

PROPERTIES:

Complete:

Time:

Space:

Optimal:



APPLICATIONS:

1.
2.

Design:

Implementation:

Result and discussion:

Learning Outcomes: Students should have the ability to

LO1: identify a problem which can be solved using uninformed search methods.

LO2: implement uninformed search methods.

LO3: describe properties of uninformed search algorithm.

LO4: identify advantage and disadvantage of the algorithm.

Course Outcomes: Upon completion of the course students will be able to evaluate various problem solving methods for an agent to find a sequence of actions to reach the goal state.

Conclusion:

Viva Questions:

1. Describe the structure of a search space in which iterative deepening search performs much worse than DFS.
2. Name which algorithm overcomes drawback of DFS and BFS?
3. Compare and contrast uninformed search strategies with respect to solving 8 puzzle problem.

For Faculty Use

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
Marks Obtained				